Shengjie Zhu

shngjz@outlook.com | 517-802-7733 | **Homepage** | **GoogleScholar** | **LinkedIn**

About me

My expertise lies in Spatial AI systems: designing customer-facing agentic front-ends grounded in a 3D-vision backend spanning camera calibration, localization, reconstruction, and Structure-from-Motion.

Work Experience

Aug, 2024 – now **Applied Scientist II, Ring AI, Amazon**.

1. Developed Ask Ring, a spatial-aware agentic system for Ring Cameras.
   - Led Intent Routing, Content Moderation, and primary contributor to Video Query Agent.
   - A robust LiDAR-Camera calibration method, friendly to inexperienced data-collection teams. [Publication-7]
   - Indoor & Outdoor Localization and Reconstruction. [Publication-1]
   - Extended to a Unitree humanoid: world-model training for package pickup and autonomous navigation.
2. Contributed to Ring Detection models, power- and memory-optimized detectors running on-device on Ring Camera SoCs.
3. Contributed to Ring Smart Video Descriptions, a light-weight real-time VLM narrating camera events.
4. Co-designed storage tiering and training-pipeline integration for Ring Parallel Computing Cluster on AWS Slurm.

May – Aug, 2024 **Research Scientist Intern, BAIR, Google**.
Foundation depth model based Structure-from-Motion for Neural Rendering and AR/VR. [Publication-2]

2021 & 2022 **Applied Scientist Intern, Amazon Device AI**.
Improve the LiDAR generated depthmap quality with epipolar geometry. [Publication-8]

Education

2017 – 2024: **PhD, Computer Science**, *Michigan State University*, East Lansing, U.S.
Dissertation: Structure and Motion Estimation from Dense Depth and Correspondence
Advisor: Prof. Xiaoming Liu

2013 – 2017: **Bachelor of Engineering, Electrical and Electronics**, *Southeast University*, Nanjing, China.

First-Authored Publications

Localization, Reconstruction, and Structure-from-Motion:

(Under Review) **[1] On Self-Improving Structure-from-Motion: A Generalizable Multi-View Pose Estimator** [Website].
Shengjie Zhu and others

3DV'26 **[2] Marginalized Bundle Adjustment: Multi-View Camera Pose from MonoDepth** [PDF, Website].
Shengjie Zhu, Ahmed Abdelkader, Mark J. Matthews, Xiaoming Liu, and Wen-Sheng Chu

ECCV'24 **[3] Revisit Self-Supervision with Local Structure-from-Motion** [PDF, Website].
Shengjie Zhu, Xiaoming Liu

CVPR'23 **[4] LightedDepth: Video Depth Estimation in light of Limited Inference View Angles** [PDF, Website].
Shengjie Zhu, Xiaoming Liu

CVPR'23 **[5] PMatch: Paired Masked Image Modeling for Dense Geometric Matching** [PDF, Website].
Shengjie Zhu, Xiaoming Liu

CVPR'20 **[6] The Edge of Depth: Explicit Constraints between Segmentation and Depth** [PDF, Website].
Shengjie Zhu, Garrick Brazil, Xiaoming Liu

Sensor Calibration:

(Under Review) **[7] Camera LiDAR Calibration in the Eye of LiDAR** [Website].
Shengjie Zhu and others

ECCV'24 **[8] Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry** [PDF, Website].
Shengjie Zhu*, Girish Chandar Ganesan*, Abhinav Kurmur, Xiaoming Liu

NeurIPS'23 **[9] Tame a Wild Camera: In-the-Wild Monocular Camera Calibration** [PDF, Website].
Shengjie Zhu, Abhinav Kurmur, Masa Hu, Xiaoming Liu

Computer Skills

Language **Python, C++, CUDA, Matlab, PyTorch, Tensorflow, CuPy, Numba**

Talk

Aug. 08, 2023 **3D Perception from Two Views**, Google Pixel Biometrics Seminar.
Feb. 05, 2024 **Structure-from-Motion Meets Self-supervised Learning**, CMU VASC Seminar. [Link]
Jul. 09, 2024 **Structure-from-Motion from Dense Learning**, Google Pixel Biometrics Seminar.

Mentored Students

Girish Chandar Ganesan (MSU), **Qucheng Peng** (UCF), **Ziyao Wang** (UMD)

Academic Service

Reviewer for **CVPR**, **ECCV**, **ICCV**, **NeurIPS**, **3DV**, **BMVC**, **ACL ARR**, **Pattern Recognition**, **IEEE T-CSVT**, **IEEE T-MM**, **JVCIR**, **DSP**.

Other Publications

CVPR'26 **[10] From 3D Pose to Prose: Biomechanics-Grounded Vision–Language Coaching** [PDF, Website].
Yuyang Ji, Yixuan Shen, Shengjie Zhu, Yu Kong, Feng Liu